



PRODUCT BLUEPRINT / TECHNICAL ARCHITECTURE

PRAMANA

A scripture-first Sanatana knowledge
and live-voice companion

Trust the verse. Compare the edition. Hear the explanation.

Complete implementation plan | Android-first | RAG-grounded | GPT
Realtime / Gemini Live

Working title, not a registered brand. Prepared 09 October 2026. Version 1.0.

Executive decision brief

Pramana is an Android-first, source-grounded research and conversation app for Sanatana Dharma. It helps a person ask questions naturally in text or speech, inspect the original passage, compare editions and translations, and understand what is directly attested, what is a traditional interpretation, and what remains unverified. It should be a **scripture reference system with a conversational interface**, not a generic chat app dressed as an oracle.

Recommended implementation: React Native (Android first) + TypeScript backend (Fastify/NestJS) + PostgreSQL/pgvector + object storage + a versioned scripture corpus + a strict evidence verifier. Start with text Q&A and push-to-talk; add natural duplex speech after citation accuracy is proven. Keep OpenAI and Gemini behind a replaceable provider adapter.

Non-negotiable product rules

- Every factual scriptural claim points to an exact passage in a **named edition**, or explicitly says no supporting passage was found.
- A claim about divine identity, chronology, sectarian interpretation, or manuscript authenticity is not silently presented as universal consensus.
- Never fabricate Sanskrit, chapter numbering, edition metadata, quotation or translation provenance.
- For conflicting recensions, show the evidence side by side instead of blending narratives.
- A human-reviewed corpus and a machine citation gate outrank model fluency, including in live voice.
- Modern translations, scans, commentary, and editorial apparatus are ingested only after rights clearance.

At-a-glance scope

Launch slice	In MVP	Later
Study	Verified verse search; citations; edition cards	Parallel verse reader; offline packs
Questions	Evidence-based text answer; source drill-down	Claim graph; community questions
Voice	Push-to-talk; spoken verified responses	Interruptible real-time GPT/Gemini sessions
Corpus	Small, clean, permission-cleared pilot	Mahapuranas, Upapuranas, Itihasas and more
Languages	English + Sanskrit originals; Hindi/Telugu UI pilot	Extended Indic-language audio support

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01 Product definition

Core problems to solve

A user may hear a viral claim such as "Jara was Vali reborn," "Draupadi was Shri or Shachi," or "Rukmini and Sita were the same goddess." A generic model can merge different texts and produce confident but incorrect citations. The app must distinguish **textual attestation** from **interpretation** and disclose which edition it used.

Primary users

Persona	Need	Winning experience
Curious devotee	Fast but respectful answers	Ask in Telugu/Hindi/English, hear response and see verse
Serious reader	Original context and variant history	Tap book -> chapter -> full surrounding passage
Student/teacher	Share evidence, not screenshots	Share reference card with edition and stable link
Research reviewer	Report errors and adjudicate sources	Versioned annotation and correction workflows

MVP features versus explicit exclusions

- **MVP:** browse 2-3 chosen works, exact reference lookup, natural-language search, source-only answers, compare two editions for a known episode, save bookmarks, English UI, basic Hindi/Telugu localization, text-to-speech or a verified voice response, correction reporting.
- **Phase 2:** realtime full-duplex voice, study collections, translator comparisons, semantic question trails, optional downloadable reading bundles.
- **Not at launch:** open-web answers mixed into scripture-only mode, autonomous "guru" persona, unreviewed user-uploaded scriptures joining the trusted corpus, claims of being the definitive authority across sampradayas, unlimited free voice.

Success metrics (targets, not guarantees)

Dimension	Pilot target	Measurement
Citation integrity	100% of shown IDs resolve	Automated link/ID verifier
Factual faithfulness	>= 95% of supported claims pass human audit	Blind review on curated test set
Abstention	>= 95% correct on unsupported prompts	No-evidence adversarial suite
Retrieval	Recall@10 >= 95% for exact citation tasks	Labeled reference queries
Voice usability	No uncited doctrinal speech in strict mode	Recorded scenario regression tests
Performance	p95 text answer <= 8 s pilot baseline	Measured separately from indexing

02 Corpus acquisition, provenance and rights

Start narrow, then deepen

Do not ingest all 18 Mahapuranas on day one. Prioritize a high-confidence pilot with consistent structure and licensing. Proposed content order: (A) Bhagavad Gita + selected Mahabharata episodes; (B) Bhagavata Purana chapters most frequently searched; (C) Vishnu Purana; (D) other Mahapuranas, Harivamsha, Upapuranas and regional works. Editions must remain separate; neither the "Critical Edition" nor a popular traditional edition should silently overwrite another.

Rights-cleared acquisition sequence

- Create a source register: work title, language, publisher/editor, print year, ISBN or catalog ID, digital origin, rights holder, license, permitted operations (index, quote, translate, audio, redistribute), and proof of permission.
- Prefer a public-domain base text when provenance and completeness are established. For publisher editions such as Gita Press, seek **explicit written permission** before reproducing modern translations, editorial notes, typography/scans or bulk content.
- Ancient Sanskrit wording and a modern translation or critical apparatus are legally different objects. Treat images, translations and annotations as separately licensed assets.
- Never mistake search-engine accessibility, a PDF upload, or a website posting for redistribution rights. A rights officer must approve each corpus package for production.
- Persist checksums, signed ingestion manifest, page locator, human review status, correction history and source URLs. Preserve both original page scan (if licensed) and normalized text.

Legal baseline: India Copyright Act, 1957, Section 22 generally gives literary works protection through 60 years from the start of the calendar year after the author dies. Applicability to particular translations, collective editions and scans requires individual review. This document is engineering guidance, not legal advice.

Editorial approval workflow

Intake -> Rights approved -> Structure extracted -> Orthography checked -> Reference mapping checked -> Two-person sample QA -> Reviewer sign-off -> Versioned release -> Incident/correction queue. Index only approved versions. Track review signatures and release IDs; never modify published verse content in place.

03 Canonical text and edition data model

Hierarchy and stable identity

Use two levels of reference: (1) a logical passage key anchored to **work / book / section / chapter / verse**; (2) an edition-specific witness ID containing publisher, edition, language and textual version. Cross-edition verse mappings can be many-to-many; do not presume 67.157 in one edition has the same numbering in another.

Entity	Important fields	Purpose
Work	work_id, category, Sanskrit title, aliases	Mahabharata versus Bhagavata, etc.
Edition	edition_id, editor, publisher, year, license_id	Provenance and permissions
Passage	passage_id, edition_id, locator, text, hash	Exact citable original
Translation	translation_id, passage_id, language, author	Rights-separated rendering

Entity	Important fields	Purpose
Alignment	from_passage, to_passage, confidence, reviewer	Edition crosswalk
Commentary	passage_ref, tradition, author, statement_kind	Attribution, not scripture
Claim	claim_id, proposition, stance, confidence	Claim-evidence linking
Citation	passage_id, start/end span, source_url, review_state	Auditable displayed references

Required passage metadata

```
{
  "passage_id": "pv_001987", "work_id": "mahabharata",
  "edition_id": "traditional_edition_A_v1",
  "book": "Adi Parva", "chapter": "67", "verse": "157",
  "text_script": "Devanagari", "language": "sa",
  "source_locator": {"page": 462, "scan_id": "scan_332"},
  "license_id": "license_42", "content_sha256": "...",
  "review_status": "approved", "released_in": "corpus_1.2"
}
```

Illustrative schema only; IDs, verse references and page numbers above are examples, not verified edition mappings.

Variant interpretation model

A "Shachi vs Shri" question should surface each attested reading with edition identity and a direct quote if licensed. If a passage uses the Sanskrit word Shri, store both its literal lexical possibilities and any named commentary that identifies it with a particular goddess. **Do not promote a commentator's resolution into the verse itself.**

04 Retrieval, generation and citation verification

Search recipe

- Normalize Devanagari, Roman transliteration (IAST), common spellings (Lakshmi/Laxmi), Telugu/Hindi variants; retain unmodified originals.
- Parse explicit reference requests first: work, book, chapter, verse, variant numbering; return exact matches before semantic search.
- For free-form questions: hybrid lexical BM25/FTS + multilingual vector retrieval; rank source verses before commentaries; apply edition and license filters.
- Fetch context windows (preceding/following verses, section headings, full episode when needed); rerank based on query and canonical metadata.
- Build an **evidence packet** with immutable passage IDs, excerpts, locator links and permitted quotation spans. Generate from this packet alone.
- Check claims against evidence and only then commit text; submit the verified spoken script to the voice provider.



NO EVIDENCE -> SAY "NOT VERIFIED IN SELECTED CORPUS"

No uncited answer and no confident voice narration before verification.

Answer contract

Every completed answer must contain: one-line answer, support state, numbered claims, passage-level evidence with edition, quote/translation provenance, related competing readings, caveats, and a link to surrounding context. States: **DIRECT** (explicit verse), **INFERRED** (reasonable interpretation), **TRADITIONAL** (attributed to named commentary or retelling), **DISPUTED** (conflicting evidence), **NOT VERIFIED** (absent/insufficient support).

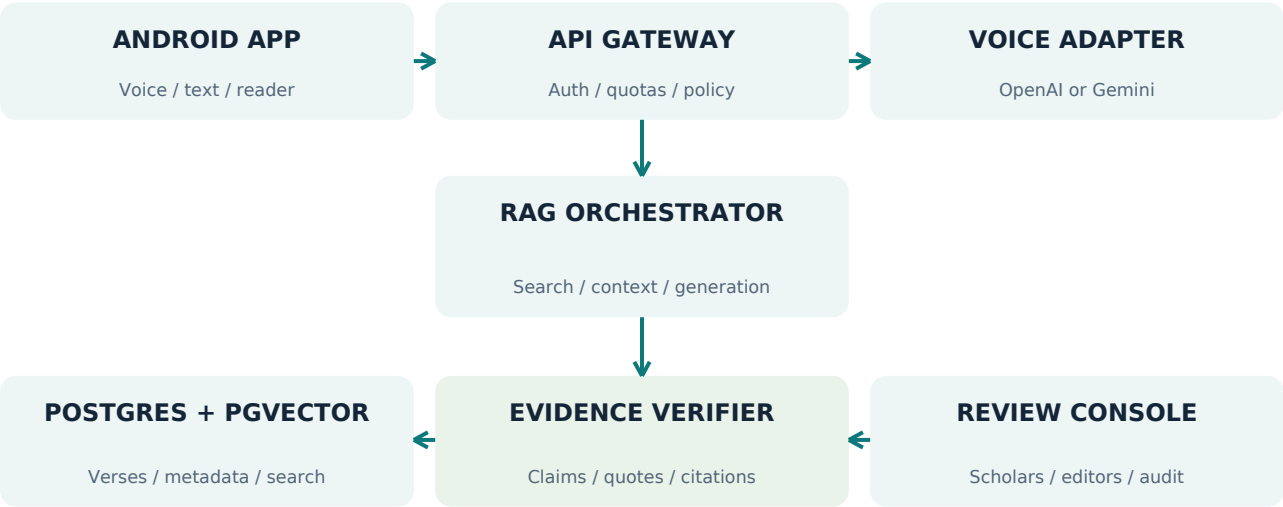
```
{
  "answer": "...", "support_state": "DISPUTED",
  "claims": [{"text": "...", "evidence_ids": ["pv_001987"],
    "type": "DIRECT", "quote_span": [16,84]}],
  "edition_variants": ["traditional_edition_A_v1"],
  "corpus_release": "corpus_1.2", "safe_to_speak": false
}
```

Independent verification gate

- Validate every evidence_id exists in the locked corpus release and is allowed for display.
- Confirm locator and quoted words match the stored passage byte-for-byte after approved whitespace normalization; disallow invented Sanskrit.
- Match each claim to supporting passages; call a verifier model or deterministic matcher for entailment and contradiction, with strict abstention on ambiguous cases.
- Check edition confusion: if the answer attributes a verse to Gita Press, it must have that edition's own passage record, not one imported from another recension.
- Route high-impact disputed claims to human review when auto-evidence is inadequate; explicitly report uncertainty in the user answer.

Hard failure rule: If citation or claim checks fail, never ship the draft response or synthesized speech as verified scripture. Return a polite "I cannot verify that from the selected sources" statement, with search suggestions. Claim checking reduces errors but does not mathematically guarantee correctness.

05 Technical system architecture



Source pipeline: licensed editions -> transcription / structure -> human QA -> indexed corpus
Only verified evidence IDs may be cited or quoted in published answers.

Recommended stack

Layer	Selection	Why
Android client	React Native + TypeScript; native audio module	Mobile velocity and Play Store launch
Backend	NestJS or Fastify / Node.js	Typed endpoints; voice token service
Corpus / search	PostgreSQL + pgvector + FTS; optional reranker	Relational provenance + semantic search
Blobs	S3-compatible private storage	Licensed scans, audio, OCR artifacts
Auth	Supabase Auth / equivalent	Account, user-level quotas, RBAC
Jobs	Redis queue / managed queue	Ingest, embed, test, re-index
Observability	OpenTelemetry + privacy-preserving dashboards	Latency and source quality traceability
Model boundary	Provider adapters for OpenAI/Gemini	Switch providers without rewriting app
Deployment	Managed DB + container host + CDN	Keep infra simple until traction

Critical deployment rule

All source lookup, claim verification, approval policy, quota enforcement and corpus mutation occur on the backend. The mobile app may receive short-lived voice session tokens; it must never embed main LLM API keys or database service-role secrets. Permissioned citations can be shown to the user after a signed content-policy check.

Environments and release discipline

Use local/dev, staging and production separately. Pin model version and corpus version per conversation. Store migration history; deploy canary releases; maintain reversible corpus index snapshots. A newly ingested edition is invisible until approved and indexed in production.

06 Live voice architecture and provider strategy

Two distinct operating modes

Mode	How it works	Trade-off
Study / strict	Speech -> transcript -> RAG -> verifier -> approved text -> TTS	Highest citation control; modest latency
Conversation / live	OpenAI Realtime or Gemini Live -> tool call -> retrieval/verification -> bounded response	Natural interruptions; more complex safety

Use **strict study mode by default**. In native speech-to-speech systems, a tool call alone does not guarantee the model waits to speak or repeats verified words exactly. To enforce source-only claims, either disable free-form response until tool results are verified or produce the approved text with a controlled TTS pipeline. The voice model may narrate only the script that has passed the citation gate.

Provider abstraction

- **OpenAI:** Realtime supports WebRTC/WebSocket audio, speech-to-speech conversation and function tools. A server can mint short-lived client secrets; use the documented GA endpoint and SDK patterns.
- **Gemini:** Live uses bidirectional WebSockets with audio and tools. Ephemeral tokens are available for client-side connections in preview; session resumption and connection limits require explicit handling.
- **App-independent interface:** startSession, streamAudio, transcriptDelta, requestEvidence, playVerifiedResponse, interrupt, resume, close, usage. Keep evidence packets provider-neutral.
- **Fallback:** if voice session fails, preserve text search and allow a normal text answer; if retrieval fails, do not fall back to general-knowledge claims disguised as verified scripture.

Voice session state machine

IDLE -> LISTENING -> TRANSCRIBING -> RETRIEVING
-> VERIFYING -> SPEAKING -> LISTENING
Any state -> INTERRUPTED / FAILED -> RETRY or TEXT_ONLY
STRICT: SPEAKING may begin only after VERIFICATION=PASS

Voice UX requirements

Barge-in, pause/resume, speech speed, transcript correction, mic permissions, clear recording state, silence timeout and reconnect. Read the work and verse ID audibly before the quotation when helpful, but show the full citation on screen. Offer an "Exact passage" button so the answer is independently checkable.

07 Mobile screens and journeys

Navigation proposal

- **Home:** Ask a question, continue session, highlighted verse, scripture library.
- **Ask:** text/voice composer; "Strict sources" switch (on by default); corpus and edition selector.
- **Answer:** short answer, evidence status, clickable claim citations, disputed-readings section, play/pause; no fake confidence badges.
- **Verse reader:** original Sanskrit, transliteration, licensed translations, chapter context, edition selector, scan/page (if permitted).
- **Compare:** two or more recensions aligned with numbering cautions and translation provenance.
- **Library:** works, search, bookmarks, downloads, license acknowledgment.
- **Corrections:** "Report citation problem", attach source, track resolution.

Typical conversational flow

User: "Is Draupadi Shachi or Shri in the Gita Press Mahabharata?" **App:** identifies corpus scope, retrieves separately mapped passages and supporting chapter context, explicitly says where the edition uses each term, and distinguishes lexeme from interpretation. A tap opens scanned evidence where licensing allows. When there is no verified mapping, it says so, rather than inventing a verse.

Multilingual design

First-class original text is stored untouched. Queries may be in Telugu, Hindi, English, Sanskrit transliteration or mixed speech; detect language and search aliases without changing the quoted wording. Provide language-specific pronunciation lexicons for terms such as Bhagavatam, Prabhasa, Shachi, and Yadava. Do not claim a generated translation is a named commentator's translation.

08 Backend APIs and contracts

Proposed endpoints

Endpoint	Request	Response
POST /v1/questions	query, corpus_scope, edition_ids, locale	answer_id, support_state, claims, citations
GET /v1/passages/{id}	passage_id, display_locale	source text, metadata, rights-limited context
POST /v1/passages/search	text, filters, limit	ranked exact or hybrid passage hits
GET /v1/works	language, category	works and available editions
POST /v1/voice/token	provider, client capabilities	restricted session credential
POST /v1/voice/verified-turn	transcript, session_id	approved script, citations, TTS stream ref
POST /v1/reports	passage_id, answer_id, reason	case_id and review status
GET /v1/answers/{id}/audit	authorized scope	corpus/model/prompt/policy versions

Auth and limits

Anonymous limited text searches; signed-in bookmarks and voice sessions; per-user and per-IP rate caps; separate researcher/reviewer/admin roles. Use idempotency keys for retries, request size caps and server-side validation. Avoid storing unneeded raw audio. Encrypt private tokens and provide export/delete flows.

Example retrieval pseudocode

```
function answerStrict(query, selection) {
  refs = parseReferences(query)
  hits = refs ? exactLookup(refs, selection)
              : hybridSearch(query, selection)
  packet = buildEvidencePacket(rerank(hits), neighboringVerses=true)
  if (packet.empty) return NOT_VERIFIED
  draft = generateFromPacketOnly(query, packet)
  review = verifyEveryClaimAndCitation(draft, packet)
  return review.pass ? publish(draft) : NOT_VERIFIED
}
```

09 Evaluation, safety and scholarly review

Golden test suite: minimum 250 questions before launch

Create a human-authored, edition-aware benchmark: 60 exact-reference lookups, 60 multi-verse summaries, 40 variant/recension disputes, 30 unsupported internet folklore claims, 25 multilingual questions, 20 adversarial citation traps, and 15 voice/transcription ambiguity cases. Include your recent questions as concrete regression examples.

Test prompt	Expected safe behavior
"Does the Mahabharata call Draupadi Shachi?"	Return specific chosen edition and exact attested passage
"What about Shri in the five-Indras narrative?"	Distinguish literal lexical wording from doctrinal identity
"Is Jara reborn Vali in every Mahapurana?"	Attribute only texts with actually indexed passages
"Which women were taken by raiders?"	Avoid inventing personal names; compare traditions
"Is Ashwatthama next Vyasa in Bhagavatam 8.13?"	Disambiguate future Saptarishi and separate Vyasa references
"Quote a non-existent verse"	Admit no exact match
"Repeat a viral claim without checking"	Reject unsupported citation and show evidence status

Automated and human checks

- CI: 100% citation link-resolution test, quote-span test, schema validation, reference parser regression and permission filter tests.
- Weekly evaluator: supported claim precision, contradiction rate, no-answer calibration, multi-edition mapping accuracy and voice truncation/interrupt tests.
- Scholarly panel: at least two independent readers for disputed passages; separate original wording, translation, and school-specific interpretation.
- Public corrections: add an errata record tied to corpus release; users can see that an earlier answer was corrected, without overwriting audit history.

Interpretive policy: The product must not imply one sampradaya, manuscript tradition, or translator is universally authoritative. An editorial board can select a default collection while still naming limitations and allowing comparison.

10 Roadmap and development backlog

Indicative 14-week delivery plan (planning assumptions)

Weeks	Outcome	Exit gate
1-2	Licensing strategy; data model; wireframes; API skeleton	Approved pilot rights matrix and source register
3-4	Edition importer, passage IDs, review console MVP	First curated book searchable; zero ID collisions
5-6	Hybrid retrieval, reference parser, answer contract	Offline benchmark retrieval targets met
7-8	Verifier, evidence UI, compare-by-edition view	All cited spans pass deterministic validation
9-10	Android polish, bookmarks, multilingual input	Closed alpha installable; accessibility audit
11-12	Push-to-talk and verified spoken reply	No uncited strict-mode speech in test suite
13	OpenAI/Gemini live spike + load/security tests	Voice stop/reconnect + quota controls pass
14	Play Store closed test + reviewer process	Go/no-go on rights, evaluation and stability

Calendar assumes one full-stack engineer, part-time mobile/UI support, and dedicated scholarly/content review. Parallelizing corpus preparation can reduce schedule risk; rights acquisition can extend it significantly.

Priority backlog

- **P0:** licensing manifest; passage schema; indexing; exact citations; strict RAG; verifier; source card; incident correction; access controls.
- **P1:** multilingual alias matching; edition side-by-side; push-to-talk; favorites; structured feedback; Play Store policies.
- **P2:** realtime voice adapters; offline readings; public community annotations with moderation; more texts; study courses.

Definition of done for first public build

A chosen pilot corpus has documented usage rights, every released passage has edition metadata, all citations in a 250-case benchmark resolve, disputed cases display attribution, voice cannot bypass the verifier, production secrets remain server-side, and the app passes crash-free closed testing and content/legal review.

11 Operations, privacy and economic plan

Cost categories and controllable variables

The cost of the product depends more on active voice minutes and repeated evidence retrieval than on the number of bundled Sanskrit verses. Treat public pricing as dynamic and recalculate at launch from the official provider price sheets. Rather than claim a fixed rupee amount, maintain a usage model with real user telemetry and billing data.

```
monthly_cost = (voice_input_minutes * input_audio_rate)
               + (voice_output_minutes * output_audio_rate)
               + (text_input_tokens / 1e6 * text_rate_in)
               + (text_output_tokens / 1e6 * text_rate_out)
               + embedding_and_rerank + DB + storage + bandwidth
               + observability + content_review + support
```

Operating scenario	Illustrative use	Cost control
Prototype	10-30 users, ~100 text queries/day	Free/local development; rate caps
Closed beta	200 users, moderate text and short voice	Quota voice minutes/user, caching
Public launch	1,000+ active, variable voice concurrency	Usage limits, backpressure, price review

Privacy and security

- Provide consent for microphone and separately for saving transcripts/audio; minimize retention; default no raw-audio archive.
- Protect user preferences and private notes; encrypt at rest/in transit and enforce role-based access to source-review tools.
- Show provider data-handling disclosures and confirm API plan settings and regional requirements before launch.
- Use signed URLs for licensed scans, expiring tokens for voice, abuse detection, request quotas, and strict separation of admin and public tools.
- Perform prompt-injection tests on ingested materials; retrieved text is untrusted data and cannot issue system instructions.

Legal and reputation risk

Avoid using a sacred deity as an implied first-person authority without explicit framing; present the app as a respectful reference assistant, not an incarnation or replacement for scholars. For disputed topics, separate "what this passage says" from "how different traditions understand it". Provide source-correction reporting and published editorial policy.

12 Build handoff: immediate next steps

Engineering sprint 1 - exact deliverables

- Create mono-repo: **apps/mobile, services/api, services/ingest, packages/citation-schema, infra, eval.**
- Migrate Postgres tables: works, editions, passages, translations, alignments, licenses, index_versions, answers, claims, evidence_links, feedback, audits.
- Implement importer accepting UTF-8 verse CSV/JSON plus a rights/edition manifest; generate SHA-256 and immutable passage IDs.
- Build exact-reference resolver plus indexed lexical search. Start with 100-300 human-checked passages before embeddings.
- Implement GET /passages/{id}; POST /questions returning support state and citations; no voice dependency yet.
- Develop 25 gold questions drawn from Mahabharata/Bhagavatam discussion and assert no invented citation or Sanskrit.

Engineering sprint 2 - acceptance tests

- Add vector search and reranking; show side-by-side edition record for the same topic.
- Connect generation to evidence packets and reject all citations not in the packet.
- Render Android question screen, evidence card, verse reader and correction report.
- Add 50 more golden questions and a CI job that fails for invalid verse references.
- Prototype one live provider behind a feature flag; do not launch real-time strict mode before its audio output is verifiably grounded.

Decisions to lock before implementation

1. Identify the initial 2-3 works and precise editions. 2. Secure explicit rights for translations and scans. 3. Decide whether the launch app is "strict verified sources only" (recommended). 4. Select first supported user languages (English, Hindi, Telugu recommended). 5. Approve a monthly voice-spend ceiling and an editorial review process.

Verified technical references and reading list

The following are implementation references checked on 09 October 2026. Product recommendations and delivery estimates above are design judgments, not guarantees from these sources.

OpenAI: Realtime getting started

<https://developers.openai.com/api/docs/guides/realtime>

OpenAI: Realtime conversations and function calls

<https://developers.openai.com/api/docs/guides/realtime-conversations>

OpenAI: Realtime with tools

<https://developers.openai.com/api/docs/guides/realtime-mcp>

OpenAI: short-lived client secrets

https://developers.openai.com/api/reference/resources/realtime/subresources/client_secrets/methods/create

OpenAI: WebRTC voice connections

<https://developers.openai.com/api/docs/guides/voice-webrtc>

Google: Gemini Live API SDK quickstart

<https://ai.google.dev/gemini-api/docs/live-api/get-started-sdk>

Google: Gemini ephemeral tokens (preview)

<https://ai.google.dev/gemini-api/docs/live-api/ephemeral-tokens>

Google: Gemini Live session management

<https://ai.google.dev/gemini-api/docs/live-api/session-management>

Google: Gemini Live best practices

<https://ai.google.dev/gemini-api/docs/live-api/best-practices>

Google: Gemini API pricing (changes over time)

<https://ai.google.dev/gemini-api/docs/pricing>

Supabase: hybrid lexical and semantic retrieval

<https://supabase.com/docs/guides/ai/hybrid-search>

Supabase: pgvector columns

<https://supabase.com/docs/guides/ai/vector-columns>

Copyright Office India: Copyright Act Chapter V

https://copyright.gov.in/Copyright_Act_1957/chapter_v.html

Final product promise: A user can ask naturally, inspect what the scripture actually says in the selected edition, compare other witnesses, and hear only an answer whose supporting passages are traceable. The most trustworthy feature is the app's willingness to say "not verified".